HHP Interpolation and Machine-Learning Correction
Technical Schematic Report

Prepared from the current codebase in `/home/suramya/HHP-Prediction/OHC`

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Contents

Executive Summary

This report documents the current Ocean Heat Content / Tropical Cyclone Heat Potential (OHC/TCHP) workflow used in the HHP project. It explains the TEOS-10 physics calculations, the exact-location Argo–RTOFS collocation design, the interpolation methods and masking rules used for seasonal maps, and the residual-learning machine-learning pipeline that corrects RTOFS toward Argo.

The key scientific point is that raw Argo observations are globally distributed, but valid D_{26} and TCHP values only exist where the upper ocean crosses the 26°C isotherm. The key computational point is that the current strongest correction model is a year-holdout XGBoost residual model trained on 2024 collocated rows and evaluated on 2025 collocated rows.

Chapter 1

Code Inventory and Project Structure

1.1 Files of Record

File	Role in workflow
`OHC/teos_ohc.py`	Core TEOS-10 / GSW computation for D_{26}, OHC, and TCHP.
`OHC/seasonal_map_common.py`	Shared seasonal mapping utilities, interpolation kernels, support masking, and global grid construction.
`OHC/build_rtofs_at_argo_points_surface_diff_2024.py`	2024 collocation and seasonal map production for Argo, RTOFS-at-Argo, and difference-of-surfaces.
`OHC/build_rtofs_at_argo_points_multiyear.py`	Multiyear collocation table builder for machine learning.
`OHC/benchmark_rtofs_argo_tabular_models.py`	First benchmark pass over baseline, classical ML, MLP, and XGBoost models.
`OHC/run_year_holdout_xgb_2024_2025.py`	Official year-holdout XGBoost experiment: train on 2024, test on 2025.
`OHC/run_year_holdout_xgb_sweep_2024_2025.py`	Feature-set and hyperparameter sweep for the year-holdout strategy.
`OHC/evaluate_year_holdout_xgb_sweep_best.py`	Refit the best sweep models and generate grouped 2025 metrics and prediction tables.
`OHC/render_ml_corrected_rtofs_2025.py`	Render 2025 raw, corrected, and difference maps for winter and summer.
`OHC/fit_locked_xgb_feature_importance.py`	Feature importance and SHAP-style attribution for the locked XGBoost model.

1.2 Workflow schematic

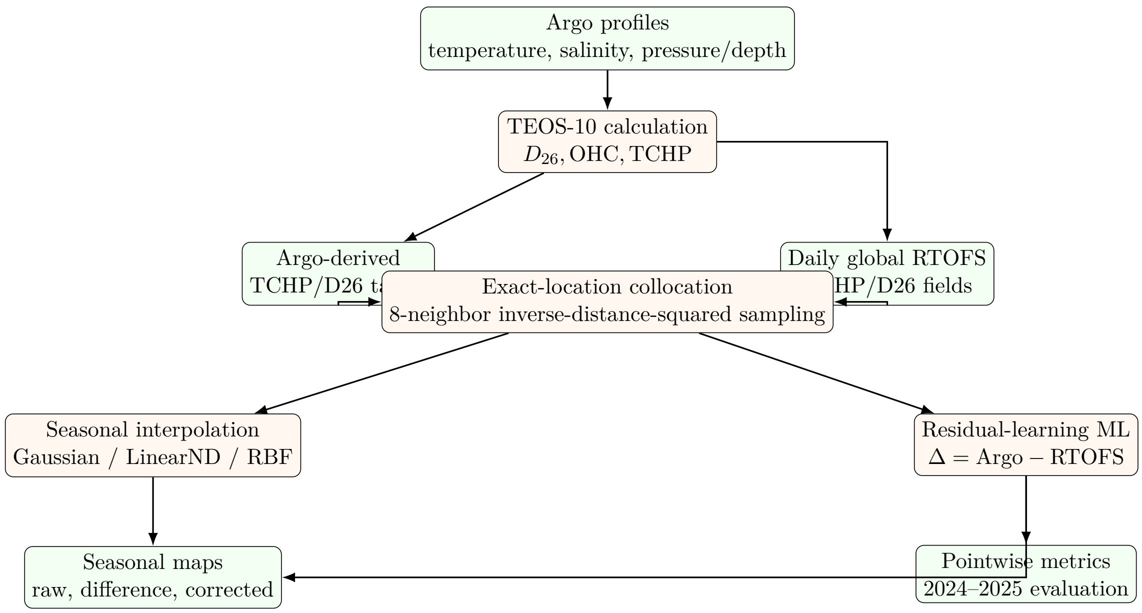

Figure 1.1: High-level HHP code schematic.

Chapter 2

TEOS-10 Calculation for D_{26}, OHC, and TCHP

2.1 Core package dependencies

The thermodynamic calculation is implemented in `OHC/teos_ohc.py`. The main numerical dependencies are:

- `gsw`: Gibbs SeaWater / TEOS-10 functions.
- `numpy`: profile cleaning, interpolation, masking, and vertical integration.

2.2 Input conventions

- `vertical`: pressure in dbar for Argo-style profiles or depth in meters for RTOFS-style profiles.
- `temp_c`: in-situ temperature profile in degrees Celsius.
- `salinity_psu`: practical salinity profile.
- `lat, lon`: required by TEOS-10 conversions.

2.3 Reference constants

$$T_{\mathrm{ref}} = 26^{\circ}\mathrm{C} \tag{2.1}$$

$$\mathrm{J\,m^{-2}} \to \mathrm{kJ\,cm^{-2}} = 10^{-7} \tag{2.2}$$

These are encoded in the code as:

```
REF_TEMP_C = 26.0
J_PER_M2_TO_KJ_PER_CM2 = 1.0 / 1.0e7
```

2.4 Mathematical definition

2.4.1 Depth of the 26-degree isotherm

The code computes D_{26} by scanning downward through the sorted profile. If the surface temperature is already below 26°C, the function returns no valid crossing.

For the first vertical pair (z_1, T_1), (z_2, T_2) such that $T_1 > 26$°C and $T_2 \leq 26$°C, the crossing depth is linearly interpolated:

$$D_{26} = z_1 + \frac{T_1 - 26}{T_1 - T_2}(z_2 - z_1). \tag{2.3}$$

2.4.2 TEOS-10 heat-content integration

Once the profile is truncated and interpolated to the exact D_{26} crossing, the code converts practical salinity to absolute salinity and computes exact density and heat capacity:

$$S_A = \text{SA\_from\_SP}(S_P, p, \lambda, \phi), \tag{2.4}$$

$$\rho = \rho_t^{\text{exact}}(S_A, T, p), \tag{2.5}$$

$$C_p = C_p^{\text{exact}}(S_A, T, p). \tag{2.6}$$

The volumetric heat excess above 26°C is:

$$q(z) = \max(T(z) - 26, 0)\,\rho(z)\,C_p(z). \tag{2.7}$$

The integrated ocean heat content above the isotherm is:

$$\text{OHC} = \int_0^{D_{26}} q(z)\,dz. \tag{2.8}$$

The tropical cyclone heat potential is then:

$$\text{TCHP} = \text{OHC} \times 10^{-7} \quad [\text{kJ}\,\text{cm}^{-2}]. \tag{2.9}$$

2.5 Pseudocode for the main computation

Algorithm 1 TEOS-10 OHC/TCHP computation

Require: vertical coordinate, temperature profile, practical salinity, latitude, longitude

Remove non-finite values and sort the profile by depth/pressure
if fewer than 2 valid levels **then**
    return no result
end if
if input axis is pressure **then**
    convert pressure to depth using `gsw.z_from_p`
else
    convert depth to pressure using `gsw.p_from_z`
end if
Find D_{26} using downward crossing of 26°C
if surface temperature is below 26°C or no crossing exists **then**
    return `d26=None, tchp=None`
end if
Interpolate temperature, salinity, and pressure to the exact D_{26} depth
Convert practical salinity to absolute salinity with `gsw.SA_from_SP`
Compute exact density and heat capacity with `gsw.rho_t_exact` and `gsw.cp_t_exact`
Compute heat excess $q(z) = \max(T - 26, 0)\rho C_p$
Integrate $q(z)$ from surface to D_{26} using trapezoidal integration
Convert J/m^2 to kJ/cm^2
return D_{26}, OHC, TCHP

Chapter 3

Why the Maps Concentrate in Warm-Water Regions

Raw Argo coverage is global, but valid D_{26} and TCHP coverage is not. The physics-based filter is the 26°C threshold.

3.1 Observed evidence from the 2024 Argo subset

Subset	Raw Argo points	Raw points with $\lvert\phi\rvert > 45°$	Valid D26/TCHP points	Valid high-latitude
Winter (JFM)	38,693	10,804	11,133	
Summer (JAS)	41,804	11,456	14,277	

Table 3.1: Raw Argo is global, but valid warm-water D_{26}/TCHP points vanish at high latitude.

This means the equatorial and subtropical appearance of the final fields is a consequence of the target definition, not a consequence of missing Argo observations at higher latitude.

Chapter 4

Argo–RTOFS Collocation and Seasonal Interpolation

4.1 Exact-location sampling rationale

The code intentionally avoids subtracting two independently gridded global surfaces directly. Instead, it first samples RTOFS at the exact Argo locations and dates. This keeps the statistical comparison tied to a shared point set.

4.2 Native-grid neighbor sampling

The collocation logic lives in two scripts:

- `OHC/build_rtofs_at_argo_points_surface_diff_2024.py`
- `OHC/build_rtofs_at_argo_points_multiyear.py`

The key parameters are:

Parameter	Value	Role
`K_NEIGHBORS`	8	Number of nearest native RTOFS cells used for exact-location interpolation
Distance space	unit-sphere xyz	Enables `cKDTree` nearest-neighbor lookup on the globe
Weighting	$1/d^2$	Inverse-distance-squared weighting for non-exact points
Exact threshold	$d \leq 10^{-6}$	If a point falls essentially on a grid cell, use that exact cell value

4.2.1 Distance conversion used in code

Lat/lon are mapped to a unit sphere:

$$\mathbf{x}(\phi, \lambda) = \begin{bmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{bmatrix}. \tag{4.1}$$

For a unit-sphere chord distance c, the great-circle distance in kilometers is:

$$d = R_\oplus \cdot 2\arcsin\left(\frac{c}{2}\right), \qquad R_\oplus = 6371\,\text{km}. \tag{4.2}$$

4.3 Pseudocode for RTOFS-at-Argo collocation

Algorithm 2 Collocate RTOFS fields to exact Argo points

Require: Argo table with date, latitude, longitude, target values
Require: Daily RTOFS TCHP/D26 fields on native model grid
Build a `cKDTree` from native RTOFS latitude/longitude converted to xyz coordinates
for each Argo point **do**
  Query the 8 nearest native RTOFS cells
  Convert chord distances to great-circle distances in km
  if an exact or nearly exact grid hit exists **then**
    assign the exact grid value
  else
    assign weighted mean using weights $w_i = 1/d_i^2$
  end if
end for
Compute residuals: $\Delta_{\mathrm{TCHP}} = \mathrm{TCHP}_{\mathrm{Argo}} - \mathrm{TCHP}_{\mathrm{RTOFS}}$, likewise for D_{26}
Save collocated rows and derived fields

4.4 Surface interpolation methods tested

The shared interpolation kernels live in `OHC/seasonal_map_common.py`.

Method	Implementation	Main parameters
Gaussian	`gaussian_interpolate()`	$L = 100\,\mathrm{km},\ R = 300\,\mathrm{km}$
LinearND	`LinearNDInterpolator`	optional support mask
RBF Gaussian	`RBFInterpolator`	$\epsilon = 100\,\mathrm{km},\ k = 64,\ s = 10^{-6}$

4.5 The 100 km support mask

The key map parameters used in the seasonal surface scripts are:

Parameter	**Value**	**Meaning**
`GRID_RES_DEG`	0.25	Rendered global grid resolution
`MASK_DISTANCE_KM`	100.0	Support mask distance from nearest collocated observation
`length_scale_km`	100.0	Gaussian smoothing scale
`truncation_radius_km`	300.0	Gaussian neighborhood cutoff
`epsilon_km`	100.0	Gaussian RBF kernel scale

The mask logic is:

$$\text{grid node valid} \iff d_{\min}(\text{grid node}, \text{observation set}) \leq 100\,\mathrm{km}. \tag{4.3}$$

The function `chord_from_km()` converts the 100 km radius to unit-sphere chord length:

$$c_{100} = 2\sin\left(\frac{100/6371}{2}\right). \tag{4.4}$$

Important interpretation

The 100 km mask is a *support constraint*. It decides where the map is allowed to display values. It is not the same as the Gaussian length scale or RBF epsilon, which control the shape of the interpolation itself.

4.6 Difference workflows that were explored

1. **Point-difference then interpolate:** compute Argo − RTOFS at collocated points first, then interpolate the difference field.
2. **Interpolate surfaces then difference:** interpolate Argo and RTOFS-at-Argo-locations separately, then subtract the seasonal surfaces.
3. **ML-corrected point predictions then interpolate:** predict corrected RTOFS values at collocated points, then render raw, corrected, and difference maps.

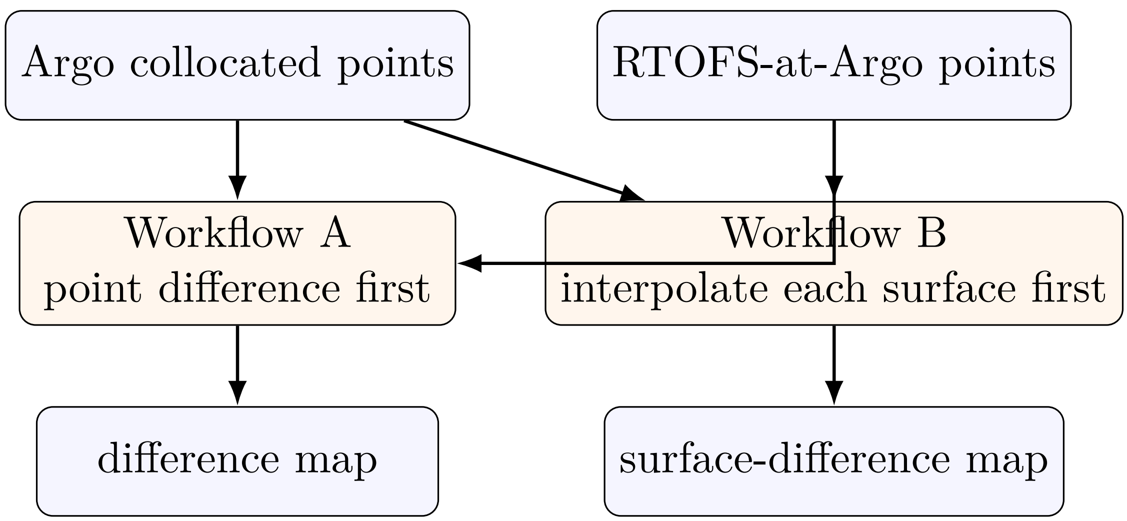

Figure 4.1: Two non-ML difference workflows explored before the ML correction stage.

Chapter 5

Machine-Learning Correction Pipeline

5.1 Training-table design

The ML stage uses the multiyear collocation table from `OHC/build_rtofs_at_argo_points_multiyear.py`. It is a pointwise table, not an image dataset.

Subset	Rows	Dates
2024 collocated rows	27,316	61
2025 collocated rows	14,031	29
Combined 2024–2025 rows	41,347	90
Finite residual rows per target	12,543	90-date universe

5.2 Residual-learning target

For each collocated point, the ML model learns a correction term rather than a direct Argo prediction:

$$\Delta_{\mathrm{TCHP}} = \mathrm{TCHP}_{\mathrm{Argo}} - \mathrm{TCHP}_{\mathrm{RTOFS}}, \tag{5.1}$$

$$\Delta_{D_{26}} = D_{26\,\mathrm{Argo}} - D_{26\,\mathrm{RTOFS}}, \tag{5.2}$$

$$\widehat{y}_{\mathrm{corrected}} = y_{\mathrm{raw\,RTOFS}} + \widehat{\Delta}. \tag{5.3}$$

5.3 First benchmark family

The first benchmark script is `OHC/benchmark_rtofs_argo_tabular_models.py`. It compares:

- raw RTOFS baseline,
- mean residual baseline,
- ridge regression,
- random forest,
- gradient boosting,
- sklearn MLP,

- XGBoost,
- simple mean ensemble of nonlinear predictors.

5.4 Leakage-aware split design

The locked protocol, frozen in `OHC/output/ml_benchmarks/locked_protocol_note.md`, uses:

- split unit: **date**,
- split type: **forward-chaining blocked validation**,
- embargo: **1 date**,
- folds: **3**,
- random row splits: **forbidden**.

The reason is to reduce leakage from same-day spatial clustering and adjacent ocean states. This keeps the evaluation closer to a realistic forward-use scenario.

5.5 Features used in the locked protocol

Feature block	**Variables**
Time encoding	`year`, `month_int`, `month_sin`, `month_cos`, `doy_sin`, `doy_cos`
Location	`lat`, `lon`, `abs_lat`
Sampling geometry	`nearest_rtofs_grid_distance_km`
Season indicator	`is_winter_jfm`, `is_summer_jas`, `is_other`
Model-state context	`model_interp_tchp_kj_per_cm2`, `model_interp_d26_m`

5.6 Pseudocode for the year-holdout ML path

Algorithm 3 2024-train / 2025-test XGBoost correction

Require: multiyear collocation table with Argo observations, raw RTOFS-at-Argo values, and residual targets

1: Keep only rows with finite Argo target, finite raw RTOFS target, and finite residual target
2: Build engineered features from date, season, location, and raw model state
3: Split rows by year: train on 2024, test on 2025
4: Fit preprocessing pipeline (median imputation; no scaling for tree models)
5: Train XGBoost regressor on the residual target Δ
6: Predict residuals on 2025 rows
7: Form corrected prediction: $\widehat{y}_{\text{corr}} = y_{\text{raw}} + \widehat{\Delta}$
8: Compute MAE, RMSE, bias, and correlation against Argo at 2025 collocated points
9: Group results by season and region

5.7 Strongest current model and exact hyperparameters

5.7.1 Locked benchmark model

The strongest practical single-model baseline is XGBoost with:

```
XGBRegressor(
    random_state=0,
    n_estimators=400,
    max_depth=5,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    reg_lambda=1.0,
    objective="reg:squarederror",
    n_jobs=8,
)
```

5.7.2 Best year-holdout sweep models

The year-holdout sweep uses the same 2024-train / 2025-test setup plus an inner blocked time-validation step. The best models found were:

- **TCHP:** enriched feature set, 3 inner folds, with $n_{\text{est}} = 300$, depth $= 4$, rate $= 0.03$, and subsample / colsample $= 0.8$.
- **D26:** base feature set, same XGBoost hyperparameters.

5.8 Feature-enrichment logic in the sweep

The sweep script adds the following beyond the base features:

- coarse region indicators: Atlantic, Indian, Pacific, northern high latitude, southern high latitude,
- warm-water availability flags,
- interaction terms such as `model_tchp_x_abs_lat` and `model_d26_x_abs_lat`.

5.9 Current benchmark performance

5.9.1 Locked forward-chaining benchmark

Target	Model	MAE	RMSE	Bias	Corr.
D26	Raw RTOFS	14.61	19.86	-7.28	0.888
D26	Locked XGBoost	10.94	14.71	-0.54	0.928
TCHP	Raw RTOFS	16.26	22.64	-8.97	0.915
TCHP	Locked XGBoost	11.86	16.73	0.39	0.944

Table 5.1: Official locked out-of-fold summary over 2024–2025 blocked folds.

5.9.2 Year-holdout train-2024 / test-2025 summary

Target	Raw RTOFS MAE	Best corrected MAE	Improvement
TCHP	16.17	11.71	4.46
D26	14.82	11.55	3.28

Table 5.2: Best 2024-train / 2025-test correction performance at collocated points.

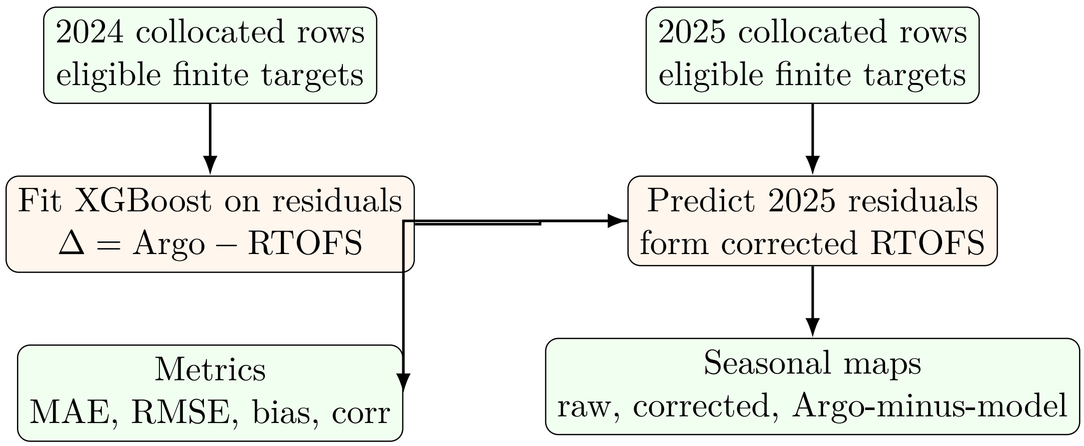

Figure 5.1: Year-holdout ML schematic used for the current strongest correction workflow.

Chapter 6

Key Interpretive Points

1. The warm-water appearance of the final D_{26}/TCHP maps is caused by the 26°C threshold and not by a lack of global Argo coverage.
2. The 100 km mask is a support mask that suppresses display where collocated observations are too far away; it is not the same as the smoothing kernel scale.
3. The current machine-learning pipeline is a point-collocated residual-learning system, not yet a gridded deep-learning system.
4. The current strongest deployable path is the year-holdout XGBoost correction trained on 2024 and tested on 2025.

Appendix: Main artifact paths

- Raw Argo distribution summary:
  `OHC/output/argo_raw_2024_distribution/summary_2024_raw_distribution.json`
- 2024 surface-difference summary:
  `OHC/output/rtofs_at_argo_points_2024_surface_diff/data/summary_2024_winter_summer.json`
- Multiyear collocation summary:
  `OHC/output/ml_collocation/data/summary_collocation_2024_2025.json`
- Locked evaluation protocol:
  `OHC/output/ml_benchmarks/locked_protocol_note.md`
- Locked XGBoost OOF summary:
  `OHC/output/ml_benchmarks/locked_xgb_oof_summary.csv`
- Year-holdout sweep best summary:
  `OHC/output/ml_benchmarks/year_holdout_xgb_sweep_best_summary.csv`